AP PRECALCULUS

Unit 1 • Lesson 1.2 • Notes (filled in)

Topic 1.2 • Rates of Change

Notes (filled in)

Learning goals

- Compute the average rate of change of a function over an interval, from a formula, a table or a graph.
- Interpret an average rate of change in context, with units.
- Estimate the rate of change at a point, and relate its sign to increasing and decreasing behavior.

Vocabulary. average rate of change, secant line, slope, rate of change at a point, units.

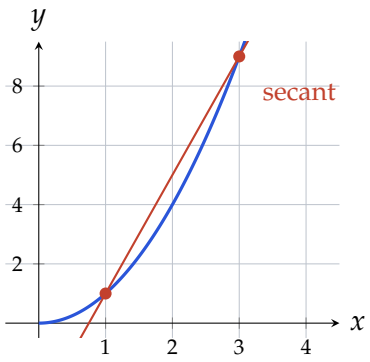
1. Average rate of change

Key idea

The average rate of change of f over the interval $[a, b]$ is

$$\frac{f(b) - f(a)}{b - a}.$$

It is the slope of the secant line through the points $(a, f(a))$ and $(b, f(b))$. Its units are output units per input unit.

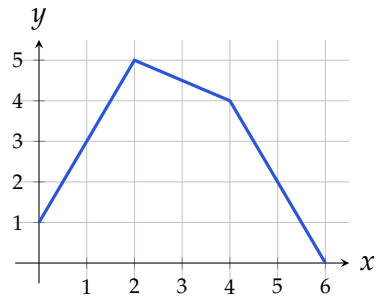


Example 1 • From a formula

Find the average rate of change of $f(x) = x^2$ on $[1, 3]$, and of $g(x) = x^3 - 4x$ on $[-2, 1]$.

$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4.$ $\quad g(1) = -3, g(-2) = -8 + 8 = 0$: $\frac{-3 - 0}{1 - (-2)} = -1.$

Example 2 • From a graph



Find the average rate of change of h on $[0,2]$, $[2,6]$ and $[0,6]$.

$[0,2]$: $\frac{5-1}{2} = 2$. $[2,6]$: $\frac{0-5}{4} = -\frac{5}{4}$. $[0,6]$: $\frac{0-1}{6} = -\frac{1}{6}$.

2. Interpreting in context

Template

"On average, [quantity] __increases or decreases__ by [number] [output units] per [input unit] from [input a] to [input b]."

Example 3 • Units

$V(t)$ is the number of visitors to a website t hours after midnight; $V(2) = 150$ and $V(6) = 870$. Compute and interpret $\frac{V(6)-V(2)}{6-2}$.

$\frac{870-150}{4} = 180$. From 2 a.m. to 6 a.m., the number of visitors increased on average by 180 visitors per hour.

3. Rate of change at a point

The rate of change at $x = a$ can be __estimated__ by average rates of change over __smaller and smaller__ intervals that contain a, using values on both sides.

Example 4 • Estimating

Estimate the rate of change of $f(x) = x^3$ at $x = 2$.

$[2,2.1]$: $\frac{9.261-8}{0.1} = 12.61$. $[2,2.01]$: 12.0601. $[1.99,2]$: $\frac{8-7.880599}{0.01} = 11.9401$.
The values approach 12: the rate of change at $x = 2$ is about 12.

Sign of the rate of change

If the rate of change is positive on an interval, f is __increasing__ there; if it is negative, f is __decreasing__. At a point where f changes from increasing to decreasing, the rate of change changes from __positive to negative__.

Common mistakes

- Subtracting in different orders: $\frac{f(b)-f(a)}{a-b}$.
- Giving the change $f(b) - f(a)$ instead of the rate.
- Forgetting units in an interpretation.

Check your understanding

1. The average rate of change of $g(x) = 5 - x^2$ on $[-1, 2]$ is -1.
2. If $T(4) = 21$ and $T(10) = 9$ (°C, hours), the average rate of change on $[4, 10]$ is -2 °C per hour.